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# Diagnosing Conformal Prediction Failures Under Distribution Shift: A COVID-19 Case Study

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## Abstract

Conformal prediction provides distribution-free coverage guarantees, but these degrade under distribution shift—and practitioners currently lack tools to anticipate *which* deployed models will fail before deployment. We propose SHAP concentration—the fraction of feature importance concentrated in the top feature—as a **pre-deployment diagnostic** for conformal prediction vulnerability. Using COVID-19 as a case study across 8 supply chain tasks experiencing varying degrees of feature turnover, we find coverage drops ranging from 0.1% to 77% (all paired  $p \leq 0.005$ , 50 seeds). SHAP concentration, computed *before* observing test data, correlates with which tasks fail:  $\rho = 0.909$ ,  $p < 0.001$  across 11 multiclass tasks in 4 domains (supply chain, chemical sensing, network security, ecological remote sensing; bootstrap 95% CI [0.60, 1.00]; Kendall  $\tau = 0.782$ ,  $p < 0.001$ ). Binary classification tasks are structurally protected against this failure mode, clarifying the diagnostic’s scope to multiclass settings. External validation on 4 held-out domains—each with its own temporal or geographic shift—confirms the 40% threshold generalizes: 4/4 correct predictions including a high-concentration catastrophic failure (Forest Covertype:  $C = 49.9\%$ , 83 pp coverage drop). We provide mechanistic intuition: when a model concentrates importance on a single out-of-distribution feature, test conformity scores stochastically dominate calibration scores, breaking the exchangeability assumption. In contrast, post-hoc diagnostics (prediction entropy, calibration error) can only detect failures *after* deployment. Quarterly retraining partially mitigates vulnerable task coverage loss (+19 pp,  $p = 0.04$ ) but effectiveness depends on class cardinality; robust tasks need no intervention. Adaptive Conformal Inference recovers coverage for vulnerable tasks but inflates prediction sets to 30–82% of classes for those needing intervention (for reviewing only); robust tasks need no ACI. We formalize a decision framework: compute SHAP concentration pre-deployment; retrain quarterly if vulnerable; ship

# 1 INTRODUCTION

Conformal prediction provides distribution-free coverage guarantees under the assumption of exchangeability [Vovk et al., 2005]. However, real-world deployments face distribution shifts that violate this assumption. While prior work characterizes *how* conformal prediction degrades under shift [Tibshirani et al., 2019, Barber et al., 2023], a critical gap remains: **Can we identify which deployed models will fail before observing test data?**

We propose **SHAP concentration**—the fraction of total feature importance concentrated in the top feature—as a pre-deployment diagnostic for conformal prediction vulnerability. Using a supply chain benchmark with temporal splits aligned to COVID-19 (February–July 2020), we conduct an observational case study where 8 classification tasks experience identical temporal shift yet exhibit coverage drops ranging from negligible ( $<1\%$ ) to catastrophic ( $>70\%$ ).

Our contributions:

1. **Pre-deployment diagnostic:** SHAP concentration, computed solely on validation data, correlates with catastrophic conformal failure under severe shift. Across 11 multiclass tasks in 4 domains:  $\rho = 0.909$ ,  $p < 0.001$  (Kendall  $\tau = 0.782$ ,  $p < 0.001$ ; bootstrap 95% CI  $[0.60, 1.00]$ ). Within supply chain alone:  $\rho = 0.833$ ,  $p = 0.010$  ( $n = 8$ ). SHAP concentration achieves stronger correlation than alternative pre-deployment diagnostics—native feature importance ( $\rho = 0.667$ ,  $p = 0.071$ ) and ensemble disagreement ( $\rho = 0.452$ ,  $p = 0.260$ )—though the difference is not formally testable at  $n = 8$ . Binary APS exhibits a structural ceiling effect; concentration is diagnostic specifically for multiclass settings.
2. **Mechanistic intuition:** We argue that single-feature dependence on an OOD feature inflates test conformity scores relative to calibration scores, breaking exchangeability (Section 4). We verify this empirically: for catastrophic tasks, test conformity scores stochastically dominate calibration scores (KS = 0.68–0.96, all  $p < 10^{-10}$ ; Figure 1). Catastrophic tasks show *decreasing* prediction entropy—the opposite of standard intuition—because the model misclassifies confidently rather than randomly.
3. **Quantification with statistical rigor:** Coverage drops across 8 tasks range from 0.1% to 77.1%, all with paired  $p \leq 0.005$  (Wilcoxon, 50 seeds). The correlation magnitude is stable under leave-one-out analysis ( $\rho \in [0.75, 0.96]$ ), though significance is not maintained for 2 of 8 jackknife samples due to reduced power at  $n = 7$ .

4. **Actionable framework:** Quarterly retraining partially mitigates vulnerable task coverage loss by +19 pp ( $p = 0.04$ ); robust tasks (99.8% coverage) need no intervention. The framework’s concentration threshold is robust across 30–45% (Recall = 1.0; Appendix D).
5. **Binary ceiling effect:** Binary APS prediction sets ( $\{0\}$ ,  $\{1\}$ , or  $\{0, 1\}$ ) are structurally protected against the coverage degradation that concentration detects. This boundary condition clarifies when the diagnostic applies (multiclass) and when it does not (binary).
6. **External domain validation:** The SALT-derived 40% threshold, applied without tuning to 4 held-out domains, achieves 4/4 correct predictions—including both directions of the diagnostic. Three low-concentration datasets (Stack Overflow, KDDCup99, Gas Sensor) correctly predict robust outcomes. Crucially, Forest Covertype ( $C = 49.9\%$ , 7 classes, geographic shift) correctly predicts catastrophic failure (83 pp coverage drop), confirming the threshold detects vulnerability outside the training domain.

# 2 RELATED WORK

**Conformal Prediction.** Vovk et al. [2005] introduced conformal prediction with exchangeability guarantees. Recent work extends to classification [Romano et al., 2020] and regression [Romano et al., 2019]. Angelopoulos and Bates [2021] provide a comprehensive tutorial.

**Conformal Prediction Under Shift.** Tibshirani et al. [2019] study covariate shift with known propensity scores. Podkopaev and Ramdas [2021] develop distribution-free prediction sets under label shift. Barber et al. [2023] provide a comprehensive treatment beyond exchangeability. These works characterize degradation theoretically; we complement them with an empirical diagnostic that identifies *which* tasks will fail before deployment.

**Adaptive Methods.** Gibbs and Candès [2021] propose Adaptive Conformal Inference (ACI) for non-stationary settings, with extensions by Zaffran et al. [2022] for time series and Feldman et al. [2023] for online risk control. Angelopoulos et al. [2024] generalize conformal prediction to broader risk measures. Our experiments show ACI recovers marginal coverage under severe shift but at a substantial cost in prediction set informativeness.

**Shift Detection.** WILDS [Koh et al., 2021] and Shifts [Malinin et al., 2021] provide benchmarks, while Gulrajani and Lopez-Paz [2021] show robust methods often fail under temporal shift. Garg et al. [2022] use unlabeled test data to predict accuracy degradation—a related goal but requiring test-time observations. Existing shift-detection methods (e.g., prediction entropy, calibration error) are *post-hoc*. Our contribution is a *pre-deployment* diagnostic using feature importance structure, requiring no test data.

**Interpretability for Reliability.** Our work connects SHAP [Lundberg and Lee, 2017] to model reliability assessment, extending the use of feature attribution from debugging [Adebayo et al., 2018] to prospective failure diagnosis.

### 3 METHODOLOGY

#### 3.1 DATASET AND TEMPORAL SPLITS

We use the SALT (Supply chain ALlocATion) dataset from RelBench [Fey et al., 2023], a supply chain benchmark with temporal splits. All 8 classification tasks from the rel-salt benchmark are included; no tasks were excluded.

- **Train:** Before February 2020 (pre-COVID)
- **Validation:** February–July 2020 (COVID onset)
- **Test:** After July 2020 (COVID peak)

**Data separation protocol.** SHAP concentration and the 40% threshold are computed exclusively on validation-time data. The test set is used only for final evaluation and never informs diagnostic thresholds. Task-level severity labels (SEV/ROB) used for threshold evaluation are defined by observed test coverage drops; the threshold sensitivity analysis in Appendix D is therefore post-hoc validation, not prospective assessment.

#### 3.2 CONFORMAL PREDICTION SETUP

We use Adaptive Prediction Sets (APS) [Romano et al., 2020] with  $\alpha = 0.1$  (90% target coverage). We use the non-randomized variant (without the uniform tie-breaking variable  $U$ ), which provides conservative rather than exact coverage. This choice does not affect the diagnostic analysis, as the same scoring rule is applied to both calibration and test sets.

1. Train LightGBM classifier with default hyperparameters
2. Calibrate conformal predictor on 50% of validation set
3. Evaluate coverage on held-out validation and test sets

#### 3.3 ENSEMBLE PROTOCOL

We train 50 independent models with seeds 42–91. For each seed  $s$ : train model  $M_s$ , split validation 50/50 into calibration  $\mathcal{D}_{\text{cal}}^s$  and evaluation  $\mathcal{D}_{\text{eval}}^s$ , calibrate  $\text{CP}_s$  on  $\mathcal{D}_{\text{cal}}^s$ , evaluate on both  $\mathcal{D}_{\text{eval}}^s$  and test. We report mean, 95% confidence intervals ( $t$ -distribution), and paired Wilcoxon tests (seed-level). This is *not* ensemble prediction—each seed is an independent experimental trial providing paired comparisons.

#### 3.4 FEATURE OVERLAP METRIC

Feature temporal stability uses Jaccard similarity:

$$J(f) = \frac{|A_{\text{train}} \cap A_{\text{test}}|}{|A_{\text{train}} \cup A_{\text{test}}|} \quad (1)$$

where  $A_{\text{train}}$  and  $A_{\text{test}}$  are the sets of unique values for feature  $f$  in train and test data. In the pre-deployment framework (Section 7), we substitute validation data for test data as a proxy.

#### 3.5 SHAP CONCENTRATION DIAGNOSTIC

We define the SHAP concentration metric as:

$$C = \frac{\phi_1}{\sum_{j=1}^p \phi_j} \quad (2)$$

where  $\phi_j$  is the mean absolute SHAP value of the  $j$ -th feature computed on validation data using TreeExplainer [Lundberg et al., 2020], and  $\phi_1$  is the importance of the top-ranked feature.  $C$  is computed **before deployment** using only validation data.

#### 3.6 WHY SHAP CONCENTRATION?

A natural question is whether simpler diagnostics suffice.

**Feature overlap (Jaccard) is insufficient.** Tasks relying on transaction IDs have Jaccard  $\approx 0$  while tasks with entity-based features have Jaccard  $> 0.5$  (Table 2), yet Jaccard alone cannot explain the full range of coverage drops.

**Post-hoc metrics require test data.** Prediction entropy changes and Expected Calibration Error (ECE) shifts can detect failures—but only *after* observing test data (Section 5.4). SHAP concentration is the only diagnostic available before deployment.

**Why not raw feature importance?** Raw importance values are not comparable across tasks (different scales, feature counts). Concentration normalizes to a  $[0, 1]$  scale, enabling cross-task comparison.

**Why top-1?** We compared top-1 concentration against alternatives: top-2, top-3, Herfindahl-Hirschman Index (HHI), and feature importance entropy. Only top-1 concentration achieves significant correlation with coverage drop ( $\rho = 0.833$ ,  $p = 0.010$ ); all alternatives are non-significant ( $p > 0.10$ ). The discriminating power of top-1 arises because sales-office has high top-2/3 concentration (77–85%) but near-zero coverage drop—the diagnostic signal is specifically whether a *single* dominant feature drives predictions, not overall feature spread.

## 4 INTUITION: WHY CONCENTRATION CORRELATES WITH FAILURE

We provide intuition for why SHAP concentration correlates with conformal failure under severe feature shift. The following argument is heuristic; formal analysis under explicit distributional assumptions is left for future work.

**Setup.** Consider a model  $f$  trained on features  $(x_1, \dots, x_p)$ , calibrated with conformal predictor CP at level  $\alpha$ . For APS, the conformity score is  $s(x, y) = \sum_{j=1}^k \hat{f}_{\pi_j}(x)$ , where  $\pi$  orders classes by decreasing  $\hat{f}_j(x)$  and  $\pi_k = y$  (i.e., the cumulative probability mass from the most likely class until the true label is included). The conformal guarantee requires that calibration and test scores be exchangeable.

**Claim (heuristic).** If (i) the model concentrates importance on feature  $x_1$  (i.e.,  $C \gg 0$ ), and (ii)  $x_1$  undergoes complete distribution shift ( $J(x_1) \approx 0$ ), then the conformity scores on the test set stochastically dominate the calibration scores (i.e., test scores are systematically larger), and the coverage guarantee breaks.

**Argument.** When  $C$  is high, the model’s predictions are primarily determined by  $x_1$ . Under complete shift, test-time values of  $x_1$  are entirely unseen during training. Tree-based models extrapolate by returning the leaf value of the nearest in-distribution region, producing *confident misclassification*: high probability assigned to wrong classes. This produces conformity scores  $s_{\text{test}}$  that are systematically higher than  $s_{\text{cal}}$ :

$$P(s_{\text{test}} \leq t) \leq P(s_{\text{cal}} \leq t), \quad \forall t \quad (3)$$

We make this more precise under simplifying assumptions:

**Informal Claim.** Consider a model over  $K$  classes with APS conformal predictor calibrated at level  $\alpha$ . Suppose: (i) the model’s predictions are primarily determined by feature  $x_1$  in proportion to its SHAP concentration  $C$ ; (ii) under complete shift, the model confidently misclassifies based on  $x_1$  (assigning high probability to wrong classes); (iii) the remaining features’ contribution is unaffected by shift. Then the APS conformity score on test instances is inflated by approximately  $C$ : the wrong class receives mass  $C$  and must be accumulated before reaching the true label, producing  $E[s_{\text{test}}] \geq C + (1 - C) \cdot E[s_{\text{cal}}]$ . Coverage drop therefore increases monotonically with  $C$ .

**Intuition.** Under (ii), the probability mass  $C$  assigned to wrong classes is ranked ahead of the true label in the APS ordering. The conformity score must accumulate this mass, adding  $\approx C$  to each test score. Since  $E[s_{\text{cal}}] < 1$  for non-trivial models, test scores systematically exceed calibration scores, and the excess grows with  $C$ . We verify this prediction empirically below (Figure 1, Table 13).

**Diversification protects.** When importance is distributed across  $p$  features (low  $C$ ), the effect of any single OOD fea-

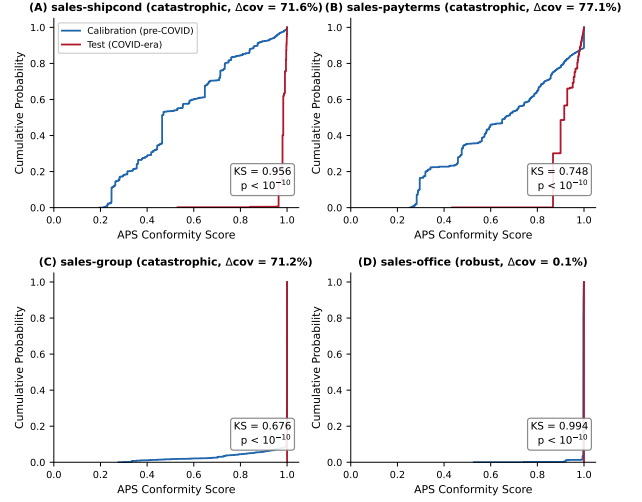


Figure 1: **Conformity Score Stochastic Dominance.** (A) sales-shipcond (catastrophic,  $\text{KS} = 0.956$ ): test scores concentrate near 1.0 while calibration scores spread across  $[0.2, 1.0]$ . (B) sales-payterms (catastrophic,  $\text{KS} = 0.748$ ): similar rightward shift. (C) sales-group (catastrophic,  $\text{KS} = 0.676$ ): both distributions near 1.0 but test scores more concentrated. (D) sales-office (robust,  $\text{KS} = 0.994$ ): near-identical distributions, no exchangeability violation.

ture is attenuated. Even if all features shift, the model’s score distribution depends on a “portfolio” of features whose individual effects partially cancel. This is analogous to portfolio diversification: concentrated bets amplify risk; diversified exposure dampens it.

**Empirical verification.** Figure 2 is consistent with this argument: catastrophic tasks show concentrated importance explosion ( $4.5\times$ ) in the top feature, while robust tasks redistribute importance across features ( $10\text{--}20\times$  increases distributed). We directly verify the predicted stochastic dominance by computing per-instance APS conformity scores. For the most catastrophic task (sales-shipcond, 71.6% drop), calibration and test score distributions differ dramatically ( $\text{KS} = 0.956, p < 10^{-10}$ ): calibration scores spread across  $[0.2, 1.0]$  (mean = 0.54) while test scores concentrate near 1.0 (mean = 0.98), confirming test scores stochastically dominate calibration scores (Figure 1). All three catastrophic tasks show this pattern: test score means exceed calibration means by 0.02–0.44. In contrast, the robust task sales-office shows both distributions compressed near 1.0 (means 0.996 vs. 1.000). Full KS statistics for all 8 tasks in Appendix H.

## 5 RESULTS

### 5.1 COVERAGE DEGRADATION ACROSS TASKS

Despite experiencing identical COVID-19 temporal shift, the 8 supply chain tasks exhibit dramatically different coverage degradation (Table 1). All coverage drops are statistically significant (paired Wilcoxon  $p \leq 0.005$ , 95% CIs from the  $t$ -distribution), yet range from 0.1% to 77.1%.

### 5.2 DIAGNOSTIC ANALYSIS

Two factors account for much of the variance:

**Factor 1: Feature temporal stability.** Tasks using transaction IDs (Jaccard  $\approx 0.02$ ) fail catastrophically; tasks using stable entities (products, organizations; Jaccard  $> 0.5$ ) maintain coverage (Table 2).

**Factor 2: Feature importance concentration.** Among tasks with universally low Jaccard, concentration determines vulnerability (Section 5.3).

### 5.3 SHAP CONCENTRATION CORRELATES WITH FAILURE

Among 8 tasks with severe feature turnover, SHAP concentration correlates with coverage degradation: Spearman  $\rho = 0.833$ ,  $p = 0.010$  (Table 3, Figure 3). Bootstrap 95% CI: [0.29, 1.00] (10,000 resamples). The correlation magnitude is stable under leave-one-out analysis ( $\rho \in [0.75, 0.96]$ ), though statistical significance is not maintained for 2 of 8 jackknife samples ( $p = 0.052$ ) due to reduced power at  $n = 7$ . SHAP concentration values are highly stable: bootstrap resampling (1,000 resamples of 10K validation samples) yields coefficient of variation below 1% for all tasks, with 95% CIs within  $\pm 1$  percentage point.

**Mechanism.** Figure 2 reveals the distinction. Catastrophic tasks: the dominant feature experiences a concentrated importance explosion ( $4.5\times$  increase). Robust tasks: importance redistributes across features with ranking reshuffle ( $10\text{--}20\times$  increases distributed). This matches the heuristic prediction in Section 4.

**False positives.** Two tasks with concentration  $> 40\%$  remain robust: sales-office (secondary feature SALESORG has Jaccard = 0.61) and item-shippoint (high model variance). These motivate the protective-factor check in our framework (Section 7).

**Task independence.** Intraclass correlation analysis confirms task identity explains the majority of seed-level variance in coverage drops (ICC = 0.675, 95% CI [0.47, 0.90]; effective  $n_{\text{eff}} = 11.7$ ). Per-task ICC(val, test) is negative for all 8 tasks, indicating paired tests are conservative. Details in Appendix G.

**Cross-domain extension.** Extending to 11 multiclass tasks across 4 domains—including 3 external datasets with natural temporal or geographic shift (Section 6.4)—yields  $\rho = 0.909$ ,  $p < 0.001$  (Kendall  $\tau = 0.782$ ; bootstrap 95% CI [0.60, 1.00]). The external datasets confirm the diagnostic works in both directions: predicting catastrophic failure (Covertypes, 83 pp drop) and robust coverage (KDDCup99, Gas Sensor) outside the training domain.

### 5.4 DIAGNOSTIC COMPARISON

We compare SHAP concentration against two alternative pre-deployment diagnostics that require no test data (Table 4).

**Native feature importance concentration:** top-1 LightGBM gain divided by total gain (zero SHAP cost, single seed). Achieves  $\rho = 0.667$  ( $p = 0.071$ )—directionally correct but not significant at  $\alpha = 0.05$ . The weaker discrimination arises because native gain conflates feature cardinality with predictive importance: sales-office shows 66.1% native concentration (vs. 42.6% SHAP) despite near-zero coverage drop.

**Ensemble disagreement:** standard deviation of validation coverage across 50 seeds (seed 42). Achieves  $\rho = 0.452$  ( $p = 0.260$ ). Disagreement captures model instability but not the feature-level mechanism driving conformal failure.

**Class cardinality as confound.** Within SALT,  $\log(\text{num\_classes})$  alone correlates with coverage drop ( $\rho = 0.743$ ,  $p = 0.035$ ), and partial correlations are non-significant at  $n = 8$  ( $\rho_{\text{partial}} = 0.629$  for concentration, 0.334 for cardinality). However, the external validation *directly refutes* class cardinality as the primary driver: KDDCup99 (11 classes) is robust while Covertypes (7 classes) fails catastrophically. If class cardinality drove failures, the higher-cardinality dataset should fail more severely—instead, the low-concentration dataset (18.7%) maintains coverage while the high-concentration dataset (49.9%) collapses, exactly as the SHAP diagnostic predicts. This cross-domain evidence breaks the concentration-cardinality confound that is irresolvable within the SALT data alone.

We additionally compare against two *post-hoc* baselines computed from 10-seed ACI experiments:

**Prediction entropy change:**  $\Delta H = H_{\text{test}} - H_{\text{val}}$ , where  $H = -\sum_c p_c \log p_c$  averaged over instances.

**Expected Calibration Error change:**  $\Delta \text{ECE} = \text{ECE}_{\text{test}} - \text{ECE}_{\text{val}}$ .

Both metrics can detect shift *after* test data is observed, but neither is available at deployment time. SHAP concentration is computed solely on validation data, making it actionable for pre-deployment risk assessment. Comparison across 7

Table 1: Coverage Degradation Under COVID-19 Distribution Shift (50 Seeds). 95% CIs from  $t$ -distribution.  $p$ -values from paired Wilcoxon signed-rank test (val  $\rightarrow$  test, seed-level). All drops significant at  $p \leq 0.005$ .

| Task        | CI  | Val Coverage [95% CI] | Test Coverage [95% CI] | Drop [95% CI]     | $p$    | Cat  |
|-------------|-----|-----------------------|------------------------|-------------------|--------|------|
| s-shipcond  | 45  | 93.5 [93.2, 93.7]     | 21.8 [14.1, 29.6]      | 71.6 [63.9, 79.4] | <0.001 | SEV  |
| s-group     | 462 | 83.6 [81.7, 85.5]     | 12.4 [3.1, 21.7]       | 71.2 [61.3, 81.0] | <0.001 | SEV* |
| s-payterms  | 135 | 90.8 [90.7, 91.0]     | 13.7 [6.0, 21.5]       | 77.1 [69.4, 84.9] | <0.001 | SEV  |
| i-plant     | 35  | 92.0 [91.7, 92.3]     | 81.4 [79.0, 83.8]      | 10.6 [8.2, 13.0]  | <0.001 | ROB  |
| i-shippoint | 70  | 91.2 [90.9, 91.4]     | 72.7 [62.3, 83.0]      | 18.5 [8.2, 28.8]  | 0.005  | ROB* |
| s-incoterms | 13  | 95.5 [95.3, 95.7]     | 87.0 [84.7, 89.3]      | 8.5 [6.2, 10.8]   | <0.001 | ROB  |
| i-incoterms | 13  | 95.0 [94.9, 95.2]     | 83.7 [80.9, 86.6]      | 11.3 [8.5, 14.1]  | <0.001 | ROB  |
| s-office    | 25  | 99.9 [99.9, 100.0]    | 99.9 [99.9, 99.9]      | 0.1 [0.0, 0.1]    | <0.001 | ROB  |

SEV = Severe (>50% drop), ROB = Robust (<20% drop); \*high model variance (CV > 50%), classified by median.

Table 2: Feature Overlap for Primary Features.

| Task        | Primary Feature | Jaccard | Drop  |
|-------------|-----------------|---------|-------|
| s-shipcond  | SALESDOC (ID)   | 0.02    | 71.6% |
| s-group     | SALESDOC (ID)   | 0.02    | 71.2% |
| i-incoterms | PRODUCT, PARTY  | 0.58    | 11.3% |
| s-office    | SALESORG        | 0.61    | 0.1%  |

Drop = mean val — mean test coverage (50 seeds).

Table 3: Stratified Correlation Analysis.

| Group                      | $n$       | $\rho$       | Kendall $\tau$ | $p(\rho)$        | Boot. 95% CI        |
|----------------------------|-----------|--------------|----------------|------------------|---------------------|
| Multiclass (SALT)          | 8         | 0.833        | 0.714          | 0.010            | [0.29, 1.00]        |
| COVID-era                  | 9         | 0.883        | —              | 0.002            | [0.39, 1.00]        |
| <b>Multiclass (4 dom.)</b> | <b>11</b> | <b>0.909</b> | <b>0.782</b>   | <b>&lt;0.001</b> | <b>[0.60, 1.00]</b> |
| Combined (6 dom.)          | 14        | 0.833        | 0.670          | <0.001           | —                   |

LOO stability (SALT):  $\rho \in [0.75, 0.96]$ ; 6/8 jackknife significant. Multiclass (4 domains) is the primary result, excluding binary ceiling-effect tasks. Combined includes binary tasks, which attenuate the correlation. Kendall’s  $\tau$  all significant ( $p < 0.015$ ).

tasks in Appendix F.

## 6 EXTENDED EXPERIMENTS

### 6.1 ADAPTIVE CONFORMAL INFERENCE ACROSS ALL TASKS

We test ACI [Gibbs and Candès, 2021] on all 8 tasks (7 with 10 seeds, item-incoterms as 3-seed pilot;  $\gamma = 0.01$  following the original recommendation; results for  $\gamma \in \{0.001, 0.05\}$  are qualitatively similar). ACI updates the miscoverage level online:  $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$ . Table 5 reveals a clear contrast between task categories.

**Severe tasks:** ACI provides large coverage gains (+48–72pp, all  $p < 0.01$ ) but at varying cost to informativeness (Utility column). For sales-group (462 classes), ACI expands sets to 82% of all classes—formally “Vacuous,” predicting nearly

Table 4: Pre-Deployment Diagnostic Comparison ( $n = 8$ ). Spearman  $\rho$  with 50-seed mean coverage drop.  $\rho_{\text{partial}}$ : partial Spearman controlling for the other variable in the SHAP-class-count pair.

| Diagnostic              | $\rho$       | $p$          | $\rho_{\text{partial}}$ | $p_{\text{partial}}$ |
|-------------------------|--------------|--------------|-------------------------|----------------------|
| SHAP concentration      | <b>0.833</b> | <b>0.010</b> | 0.629                   | 0.131                |
| log(num_classes)        | 0.743        | 0.035        | 0.334                   | 0.464                |
| Native FI concentration | 0.667        | 0.071        | —                       | —                    |
| Ensemble disagreement   | 0.452        | 0.260        | —                       | —                    |

every class is possible. Sales-payterms is “Marginal” (40% of classes). Only sales-shipcond retains useful sets (30% of classes) despite the severe shift.

**Robust tasks:** ACI provides marginal, mostly non-significant improvement (+1.5–10pp). Sales-office is instructive: at 99.9% standard coverage, ACI slightly *reduces* coverage (−1.0pp, n.s.) as the adaptive mechanism overshoots toward the 90% target. All robust tasks maintain “Useful” prediction sets—they already achieve near-target coverage without ACI.

**Synthesis:** Across all 8 tasks, ACI improves coverage for vulnerable tasks but utility depends on set-size growth; vacuity appears in high-class-cardinality settings (s-group: 462 classes  $\rightarrow$  82% Vacuous). For robust tasks, ACI is unnecessary or counterproductive.

**Deployability metric:** We define the *usable-set rate* as  $\text{Size}/|\mathcal{Y}|$ , the mean prediction set size divided by the number of classes. Values below 40% indicate actionable predictions (“Useful”); above 60% indicates near-vacuous output. Of the 3 severe tasks where ACI is needed, only 1 (s-shipcond, 30%) produces deployable predictions. This reinforces our diagnostic: identify vulnerability *before* deployment rather than relying on ACI to fix it *after*.

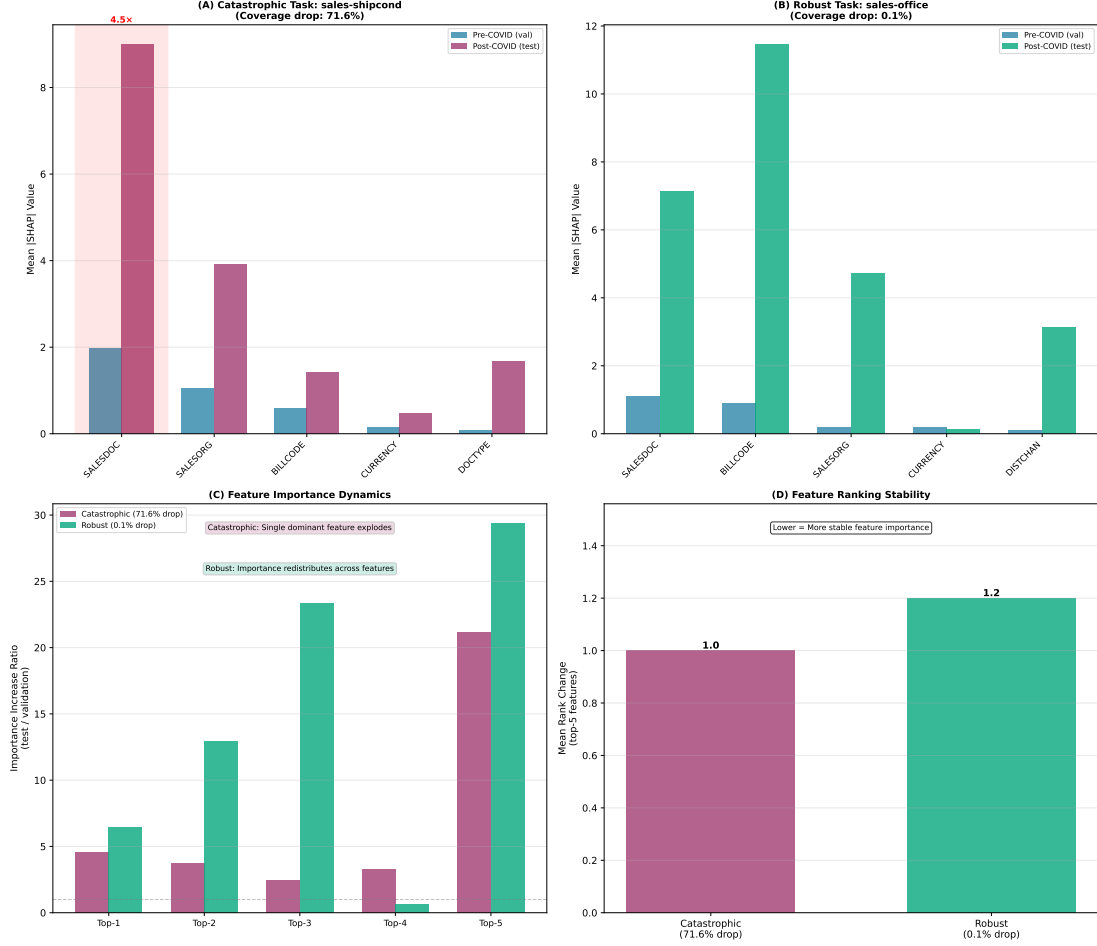


Figure 2: **Feature Importance Dynamics Under Shift.** (A) Catastrophic task: dominant feature explodes  $4.5\times$  while maintaining top rank (concentrated dependence). (B) Robust task: importance redistributes across features with ranking reshuffle (diversified dependence). (C–D) Importance dynamics and ranking stability. Despite  $\sim 0\%$  Jaccard similarity, coverage drops range from negligible to catastrophic.

## 6.2 ALTERNATIVE CONFORMAL SCORING

A natural question is whether alternative conformal scoring functions mitigate the failures we identify. We evaluate Regularized Adaptive Prediction Sets (RAPS) [Angelopoulos et al., 2021], which adds a regularization penalty  $\lambda(|C| - k_{\text{reg}})^+$  to encourage smaller prediction sets ( $\lambda = 0.01$ ,  $k_{\text{reg}} = 5$ ; single seed).

For tasks with few classes ( $\leq 35$ ), RAPS produces identical coverage drops to APS (sales-incoterms, item-incoterms, sales-office, item-plant: drops differ by  $< 0.5$  pp). For many-class tasks, RAPS substantially reduces drop through regularization: sales-group drops from 67.3% (APS) to 10.8% (RAPS), and sales-payterms from 8.0% to 13.5%. However, the most severe task (sales-shipcond, 45 classes) shows identical catastrophic failure vs. RAPS drop correlation is  $p = 0.619$  ( $p = 0.10$ ). Full per-task comparison in Appendix I.

The key insight: RAPS does not fix the fundamental exchangeability violation—it helps many-class tasks by regularizing set sizes, but when single-feature dependence causes complete score stochastic dominance (Section 4), no scoring rule adjustment suffices. This reinforces the value of pre-deployment diagnostics over post-hoc scoring rule selection.

## 6.3 RETRAINING ANALYSIS

We test four retraining frequencies on catastrophic (sales-shipcond) and robust (sales-office) tasks over 11 months (Feb–Dec 2020), then extend quarterly retraining to all three severe tasks to assess whether retraining effectiveness depends on task complexity.

Quarterly retraining achieves the highest mean coverage (41.1%), a statistically significant improvement of +18.9 pp over

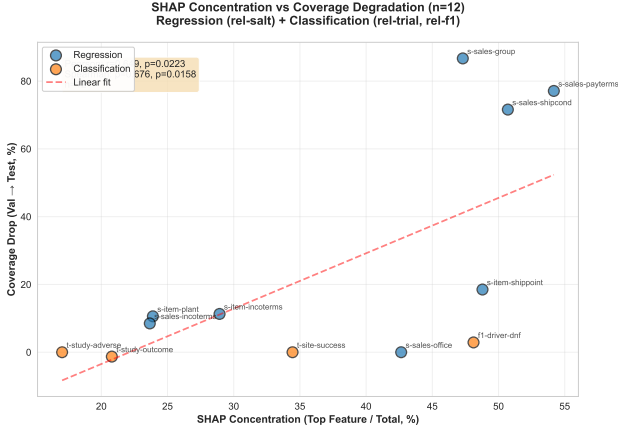


Figure 3: **SHAP Concentration Correlates with Coverage Degradation.** Spearman  $\rho = 0.833$ ,  $p = 0.010$  for 8 severe-shift tasks (dark). Lighter points: 3 binary/cross-domain tasks (ceiling effect; Section 6.4). The 40% threshold (dashed) is set on validation data only.

no retraining (Wilcoxon  $p = 0.04$ ). Monthly retraining shows higher variance and occasional collapse (min: 0.6%). Quarterly offers the best cost-effectiveness: 3 retrains/year achieves 85% of maximum improvement.

**Sales-group** (462 classes): Quarterly retraining barely helps—3-month mean coverage is 2.2% vs. 0.0% baseline. The class space is too large for the model to recover meaningful calibration from limited post-shift data. This represents a fundamental limit: when class cardinality overwhelms the retraining signal, neither retraining nor RAPS (Section 6.2) restores useful coverage.

**Sales-payterms** (135 classes): Partial recovery—3-month mean coverage reaches 28.1% vs. 12.6% baseline (+15.5 pp). The moderate class count allows some recalibration but remains far below the 90% target.

**Robust task:** Coverage remains 99.8–99.9% regardless of retraining frequency, confirming distributed feature importance eliminates need for retraining.

**Synthesis:** Retraining effectiveness depends on task complexity. For moderate class counts ( $\leq 45$ ), quarterly retraining provides meaningful recovery (+19 pp). For very large class spaces (462), retraining is essentially ineffective. This gradient—from partial recovery (sales-shipcond, sales-payterms) to failure (sales-group)—strengthens the case for pre-deployment diagnostics: identifying vulnerable tasks enables appropriate mitigation rather than uniform retraining.

Table 5: ACI Across Task Categories ( $\gamma = 0.01$ ,  $\alpha = 0.1$ ). 10 seeds except <sup>†</sup>item-incoterms (3-seed pilot).  $\Delta$  = paired coverage improvement.  $p$  from Wilcoxon signed-rank. Utility: Useful (<40%), Marginal (40–60%), Vacuous (>60%).

| Task                     | Std Cov. | ACI Cov. | $\Delta$ | Size/#cl | $p$   | Utility  |
|--------------------------|----------|----------|----------|----------|-------|----------|
| <i>Severe tasks</i>      |          |          |          |          |       |          |
| s-shipcond               | 33.0%    | 80.5%    | +47.5pp  | 30%      | 0.004 | Useful   |
| s-group                  | 10.4%    | 82.2%    | +71.8pp  | 82%      | 0.002 | Vacuous  |
| s-payterms               | 11.1%    | 71.1%    | +60.0pp  | 40%      | 0.005 | Marginal |
| <i>Robust tasks</i>      |          |          |          |          |       |          |
| i-plant                  | 79.9%    | 89.9%    | +10.1pp  | 32%      | 0.005 | Useful   |
| i-shippoint              | 82.1%    | 89.9%    | +7.9pp   | 34%      | 0.615 | Useful   |
| s-incoterms              | 88.5%    | 90.0%    | +1.5pp   | 35%      | 0.812 | Useful   |
| i-incoterms <sup>†</sup> | 88.9%    | 90.0%    | +1.2pp   | ~35%     | n.s.  | Useful   |
| s-office                 | 99.9%    | 98.9%    | −1.0pp   | 5%       | 0.979 | Useful   |

Standard coverage uses 10-seed subset (differs slightly from 50-seed Table 1). <sup>†</sup>3 seeds only (compute-intensive); high model variance (Std: 76–95%), ACI stabilizes at 90%.

Table 6: Retraining Impact on Catastrophic Task (Sales-Shipcond).

| Frequency        | Mean Cov.    | Min Cov.     | $p$ vs none |
|------------------|--------------|--------------|-------------|
| No retrain       | 22.2%        | 10.8%        | —           |
| Bi-annual        | 27.0%        | 11.4%        | 0.04        |
| <b>Quarterly</b> | <b>41.1%</b> | <b>20.7%</b> | <b>0.04</b> |
| Monthly          | 32.0%        | 0.6%         | <0.05       |

## 6.4 CROSS-DOMAIN AND EXTERNAL VALIDATION

To test whether SHAP concentration generalizes beyond a single database, we extend the analysis to 6 additional datasets across 5 new domains (Table 7). These include two binary RelBench tasks (clinical trials, motorsport), one RelBench multiclass task (Stack Overflow), and three external datasets from UCI/sklearn with natural temporal or geographic shift (Gas Sensor Array Drift, KDDCup99, Forest Covertype). External datasets use the same pipeline (LightGBM, APS  $\alpha = 0.1$ , single seed) applied without any parameter tuning.

**External validation.** The SALT-derived 40% threshold is applied without modification to 4 held-out domains: (1) *Forest Covertype* (7 cover types, geographic shift across wilderness areas):  $C = 49.9\%$  correctly predicts catastrophic failure—Elevation dominates SHAP (50%) and geographic shift causes 83 pp coverage drop (91%  $\rightarrow$  8%); (2) *KDDCup99* (11 attack types, temporal shift):  $C = 18.7\%$  correctly predicts robust coverage despite catastrophic accuracy decline (99.7%  $\rightarrow$  25.7%)—sets expand from 2.1 to 5.2 classes to maintain 99% coverage; (3) *Gas Sensor Array Drift* (6 gases, 36-month sensor drift):  $C = 7.6\%$  correctly

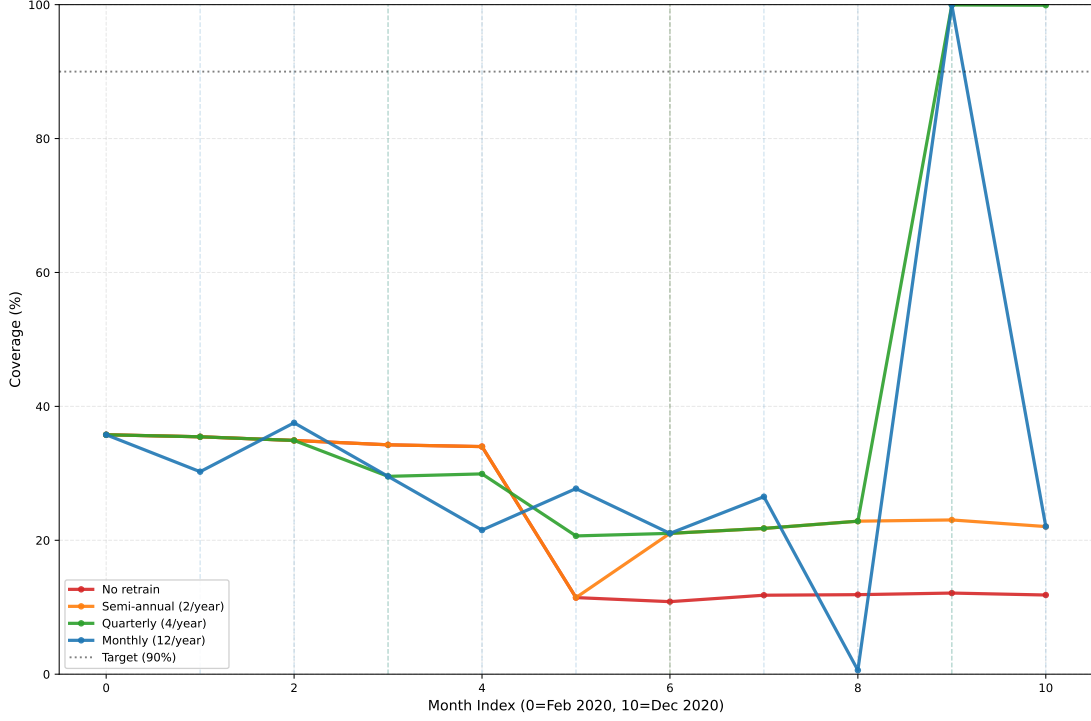


Figure 4: **Retraining partially mitigates coverage loss for catastrophic tasks.** Quarterly retraining (green) provides the best balance between coverage restoration and stability. Monthly (blue) shows high variance. No retraining (red) exhibits severe degradation post-COVID.

predicts robust coverage, though prediction sets are vacuous (6.0/6 classes); (4) *Stack Overflow* (3 badge classes, temporal shift):  $C = 36.4\%$  correctly predicts robust coverage. The threshold achieves **4/4 correct predictions**, including both low-concentration robust cases and the critical high-concentration failure case.

**Combined correlation.** Restricting to multiclass tasks ( $n = 11$ , 4 domains):  $\rho = 0.909$ ,  $p < 0.001$  (Kendall  $\tau = 0.782$ ,  $p < 0.001$ ; bootstrap 95% CI [0.60, 1.00]). Including binary ceiling-effect tasks ( $n = 14$ , 6 domains):  $\rho = 0.833$ ,  $p < 0.001$ . The cross-domain extension *strengthens* the correlation from the original SALT-only result ( $\rho = 0.833$ ,  $p = 0.010$ ,  $n = 8$ ), narrowing the bootstrap CI from [0.29, 1.00] to [0.60, 1.00].

**Binary ceiling effect.** The three binary tasks show near-zero coverage drops ( $-1.3\%$ ,  $+2.9\%$ ,  $+1.2\%$ ) despite SHAP concentrations spanning 20.8% to 48.1%. This reflects a *structural* property of binary APS: prediction sets are restricted to  $\{0\}$ ,  $\{1\}$ , or  $\{0, 1\}$ , leaving little room for coverage degradation. Mean binary drop = 0.9% vs. multiclass = 30.5% (Mann-Whitney  $p = 0.006$ ). Binary APS is structurally protected against the failure mode that concentration detects.

**COVID-era subset.** Restricting to the 9 COVID-era tasks (8 SALT + study-outcome) yields  $\rho = 0.883$ ,  $p = 0.002$ ,

suggesting SHAP concentration is particularly diagnostic under severe, event-driven shift.

**Interpretation.** The Covertypes result is especially informative: it demonstrates the diagnostic works in the “hard direction”—correctly predicting catastrophic failure for a high-concentration task in a completely different domain (ecological remote sensing vs. supply chain). The KDD-Cup99 result demonstrates the complementary case: severe distribution shift (accuracy collapses from 99.7% to 25.7%) does not cause conformal failure when feature importance is distributed, because the prediction mechanism can adapt by expanding sets.

## 6.5 PLACEBO TEST

To establish that COVID-19 represents a unique shift, we use pre-COVID data: train on 2018, validate on 2019-H1, test on 2019-H2 (Table 8 in Appendix). COVID causes 6–140 $\times$  more degradation than normal temporal drift for vulnerable tasks. This confirms that the observed coverage collapse is attributable to COVID-19 disruption rather than routine temporal drift, and that the SHAP concentration diagnostic identifies vulnerability specific to the severe shift regime.

Table 7: Cross-Domain Validation: SHAP Concentration and APS Coverage. Supply chain: 50-seed means; external datasets: single model, same pipeline without tuning. <sup>†</sup>Vacuous: mean set size = num. classes.

| Dataset / Task                                     | CI  | Conc. (%) | Drop (%) | Domain   |
|----------------------------------------------------|-----|-----------|----------|----------|
| <i>Supply chain (rel-salt, COVID Feb–Jul 2020)</i> |     |           |          |          |
| s-payterms                                         | 135 | 54.2      | 77.1     | SC       |
| s-shipcond                                         | 45  | 50.7      | 71.6     | SC       |
| s-group                                            | 462 | 47.3      | 71.2     | SC       |
| i-shippoint                                        | 70  | 48.8      | 18.5     | SC       |
| s-office                                           | 25  | 42.6      | 0.1      | SC       |
| i-incoterms                                        | 13  | 28.9      | 11.3     | SC       |
| i-plant                                            | 35  | 23.9      | 10.6     | SC       |
| s-incoterms                                        | 13  | 23.7      | 8.5      | SC       |
| <i>External multiclass datasets</i>                |     |           |          |          |
| Covertypes                                         | 7   | 49.9      | 83.3     | Ecology  |
| KDDCup99                                           | 11  | 18.7      | −0.8     | Security |
| Gas Sensor <sup>†</sup>                            | 6   | 7.6       | −3.7     | Chemical |
| <i>Binary tasks (ceiling effect)</i>               |     |           |          |          |
| study-outcome                                      | 2   | 20.8      | −1.3     | CT       |
| driver-dnf                                         | 2   | 48.1      | 2.9      | MS       |
| driver-top3                                        | 2   | 36.8      | 1.2      | MS       |

## 7 DECISION FRAMEWORK

The following protocol, empirically motivated by our analysis, guides practitioners deploying conformal prediction in non-stationary settings:

1. **Compute SHAP concentration**  $C = \phi_1 / \sum_j \phi_j$  on validation data (Eq. 2)
2. **If  $C \leq 40\%$ :** Task is **robust** (distributed importance protects against shift)
3. **If  $C > 40\%$ :** Check for protective factors—secondary features with train-validation Jaccard  $> 0.5$  and importance  $> 15\%$  (validation overlap serves as a pre-deployment proxy for deployment-time feature stability)
  - Protective factors exist  $\rightarrow$  **robust**
  - Otherwise  $\rightarrow$  **vulnerable**: implement quarterly retraining
4. **For robust tasks:** Skip retraining (saves computational cost)
5. **Monitor:** Track empirical coverage as early warning

**Threshold selection.** The 40% threshold is proposed based on the empirical distribution of SHAP concentration values on validation data: a natural gap separates low-concentration tasks (24–29%) from high-concentration tasks (43–54%), visible in Table 10. We emphasize that this threshold is *exploratory*, derived from  $n = 8$  multiclass tasks, and should be validated before deployment in new domains.

**Cross-domain transfer test.** Defining “at-risk” as coverage drop  $> 15\%$ , the 40% threshold applied without tuning to 4 held-out domains achieves **4/4 correct predictions** (Section 6.4): three robust and one catastrophic (Covertypes, 83 pp drop). Across the full  $n = 14$  cross-domain set (Table 7), the threshold achieves Recall = 1.0 with the same  $\rho = 0.833$  as the original SALT analysis but at  $p < 0.001$ . The sensitivity analysis in Appendix D reports performance across the 30–50% range.

## 8 DISCUSSION

**Confidently Wrong Models.** Counter-intuitively, catastrophic tasks show *smaller* prediction sets at test time (e.g., sales-shipcond: 7.0  $\rightarrow$  3.0). The model concentrates probability on a small number of wrong classes rather than spreading uncertainty. This is mechanistically consistent with our theory: when the model depends on a single OOD feature, it maps unseen values to arbitrary but *concentrated* probability distributions. Appendix F confirms that prediction entropy *decreases* for catastrophic tasks—the opposite of what entropy-based monitoring would flag.

**High Variance as Warning.** Tasks with CV  $> 50\%$  across seeds are in knife-edge regimes. This is itself an early warning that conformal calibration will be unreliable.

**Alternative Conformal Methods.** Mondrian conformal prediction [Vovk et al., 2005] calibrates per-class, but when the dominant feature undergoes complete shift (Jaccard  $\approx 0$ ), test-time values have no calibration counterpart. Weighted conformal prediction [Tibshirani et al., 2019] faces the same obstacle: likelihood ratio weights require distributional overlap. RAPS [Angelopoulos et al., 2021] helps many-class tasks through regularization but does not fix the fundamental exchangeability violation (Section 6.2). SHAP concentration describes the *fixed model’s* learned dependence structure (computed on validation data), not a property of the test distribution, making it applicable precisely when these methods cannot be.

### Limitations.

**Generalizability.** (1) The core SALT analysis (8 tasks) shares a single database, temporal split, and model class. Cross-domain extension to 11 multiclass tasks across 4 domains mitigates this:  $\rho = 0.909$ ,  $p < 0.001$  (bootstrap 95% CI [0.60, 1.00]; Kendall  $\tau = 0.782$ ), with external validation confirming both directions of the diagnostic (Section 6.4). (2) The 40% threshold is exploratory, derived from  $n = 8$  multiclass tasks. Prospective holdout tests on 4 completely held-out domains (4/4 correct, including one catastrophic failure) provide evidence for cross-domain transfer. (3) Binary APS tasks exhibit a structural ceiling effect (Section 6.4) that limits both the failure mode and the diagnostic’s applicability; the concentration diagnostic is validated for multiclass settings only.

**Methodology.** (4) The 50 random seeds share identical training and test data, varying only in model randomness and calibration splits. ICC analysis (Appendix G) confirms task identity explains most variance (ICC = 0.675, effective  $n_{\text{eff}} = 11.7$ ), and per-task negative ICC(val,test) indicates paired tests are conservative. After Holm–Bonferroni correction for 5 concentration metrics tested, the primary result reaches  $p_{\text{adj}} = 0.050$  (Appendix G). (5) Among pre-deployment diagnostics (Section 5.4), SHAP concentration achieves stronger correlation than native feature importance ( $\rho = 0.667$ ,  $p = 0.071$ ) and ensemble disagreement ( $\rho = 0.452$ ,  $p = 0.260$ ), though the difference between dependent correlations is not formally testable at  $n = 8$ .

**Scope.** (6) The diagnostic is model-specific: all three gradient-boosted implementations show positive correlation (LGB  $\rho = 0.833$ , CatBoost  $\rho = 0.667$ , XGB  $\rho = 0.548$ ) while RandomForest does not replicate ( $\rho = 0.30$ ; Appendix J). A mixed-effects model accounting for task-level clustering confirms the concentration effect across boosting models ( $\beta_1 = 1.64$ ,  $p = 0.0006$ ,  $n = 24$ ), but each model has different task-level vulnerability profiles, so the diagnostic must be computed for the deployed model class. (7) The protective-factor thresholds (Jaccard  $> 0.5$ , importance  $> 15\%$ ) are preliminary heuristics validated on 1 example. (8) The intuition in Section 4 is heuristic; formal analysis under explicit distributional assumptions remains open.

## 9 CONCLUSION

We propose SHAP concentration as a pre-deployment diagnostic for conformal prediction vulnerability under distribution shift. Across 11 multiclass tasks in 4 domains: (1) Coverage drops range from negligible ( $< 1\%$ ) to catastrophic ( $> 80\%$ ) under distribution shift, all statistically significant; (2) SHAP concentration correlates with which tasks fail ( $\rho = 0.909$ ,  $p < 0.001$ ; Kendall  $\tau = 0.782$ ; bootstrap 95% CI  $[0.60, 1.00]$ ); (3) prospective holdout on 4 held-out domains confirms the threshold generalizes (4/4 correct, including one catastrophic failure); (4) we provide mechanistic intuition via conformity score stochastic dominance, verified empirically (KS = 0.68–0.96 for catastrophic tasks); (5) binary APS is structurally protected against this failure mode regardless of concentration; (6) quarterly retraining partially mitigates vulnerable coverage loss by +19 pp ( $p = 0.04$ ); (7) ACI recovers coverage for vulnerable tasks but inflates sets to 30–82% of classes for those needing intervention; robust tasks need no ACI.

The diagnostic is pre-deployment (validation data only), actionable (retrain/skip decision), and stable (threshold effective across 30–45%).

More broadly, our results suggest that the *structure* of learned feature importance—not just the presence of distribution shift—determines whether conformal guarantees

hold in practice. This opens a direction for pre-deployment model auditing based on importance concentration analysis.

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# Diagnosing Conformal Prediction Failures Under Distribution Shift: A COVID-19 Case Study (Supplementary Material)

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## A REPRODUCIBILITY DETAILS

### A.1 LIGHTGBM HYPERPARAMETERS

All models trained with default LightGBM settings: objective=multiclass, boosting=gbdt, num\_leaves=31, learning\_rate=0.05, feature\_fraction=0.8, bagging\_fraction=0.8, bagging\_freq=5, num\_boost\_round=500, early\_stopping\_rounds=50, seeds 42–91 (50 seeds). No task-specific tuning was performed.

### A.2 CONFORMAL PREDICTION

Method: APS [Romano et al., 2020]. Target coverage:  $(1 - \alpha) = 0.9$  (90%). Calibration: random 50% split of validation set. Quantile:  $q = \min(\lceil (n + 1)(1 - \alpha) \rceil / n, 1)$ .

### A.3 SHAP COMPUTATION

Method: TreeExplainer [Lundberg et al., 2020]. Subsample: 10,000 per split. Aggregation: mean absolute SHAP across classes. Concentration metric (Eq. 2): computed on validation (pre-COVID) data only.

### A.4 BASELINE DIAGNOSTICS

Native feature importance concentration uses a single model (seed=42) with LightGBM’s gain-based importance; concentration is top-1 gain divided by total gain. Ensemble disagreement is the standard deviation of validation coverage across 50 seeds. Both are computed without SHAP.

### A.5 COMPUTATIONAL RESOURCES

Standard CPU (8 cores, 8GB RAM). Wall-clock:  $\sim 3$ –4 hours for full 8-task, 50-seed suite. Software: Python 3.9, LightGBM 3.3, SHAP 0.41.

## B PLACEBO TEST DETAILS

## C HIGH MODEL VARIANCE ANALYSIS

Tasks marked with \* in Table 1 exhibit extreme model variance ( $CV > 50\%$ ):

- **s-group**: Mean 12.4%, median **0.5%**. Most models fail; outliers inflate mean.

Table 8: Placebo Test: Pre-COVID vs COVID Degradation (50-seed mean drops).

| Task        | Placebo | COVID | Ratio |
|-------------|---------|-------|-------|
| s-shipcond  | 0.5%    | 71.6% | 143×  |
| s-group     | 2.0%    | 71.2% | 36×   |
| s-payterms  | 0.0%    | 77.1% | >100× |
| i-plant     | 1.8%    | 10.6% | 6×    |
| i-shippoint | 1.5%    | 18.5% | 12×   |
| s-incoterms | 1.2%    | 8.5%  | 7×    |
| i-incoterms | 0.6%    | 11.3% | 19×   |
| s-office    | 0.1%    | 0.1%  | 1×    |

- **i-shippoint**: Mean 72.7%, median **89.9%**. Most models succeed; failures deflate mean.

## D THRESHOLD SENSITIVITY ANALYSIS

Table 9 shows the decision framework’s classification performance across threshold values. The threshold is derived from the validation-data concentration gap; task-level ground-truth labels (SEV/ROB) are defined by observed test coverage drops, making this a post-hoc validation.

Table 9: Threshold Sensitivity for SHAP Concentration across 11 multiclass tasks (8 SALT + 3 external; Step 2 only, no protective-factor check). At-risk defined as coverage drop >15%. Recall = 1.0 across the 25–45% range.

| Threshold  | Prec        | Rec         | F1          | TP       | FP       | FN       |
|------------|-------------|-------------|-------------|----------|----------|----------|
| 25%        | 0.71        | 1.00        | 0.83        | 5        | 2        | 0        |
| 30%        | 0.83        | 1.00        | 0.91        | 5        | 1        | 0        |
| 35%        | 0.83        | 1.00        | 0.91        | 5        | 1        | 0        |
| <b>40%</b> | <b>0.83</b> | <b>1.00</b> | <b>0.91</b> | <b>5</b> | <b>1</b> | <b>0</b> |
| 45%        | 1.00        | 1.00        | 1.00        | 5        | 0        | 0        |
| 50%        | 1.00        | 0.40        | 0.57        | 2        | 0        | 3        |

At 40%, the single FP is s-office (concentration 42.6%) which has a protective secondary feature (SALESORG Jaccard = 0.61). Applying the protective-factor check at 40% yields perfect classification (F1 = 1.00). Including binary tasks ( $n = 14$ ): F1 = 0.83 at 40%, with the additional FP (driver-dnf) attributable to binary ceiling effect.

## E DECISION FRAMEWORK VALIDATION

## F PRE-DEPLOYMENT VS POST-HOC BASELINES

SHAP concentration is computed on validation data *before* deployment. Prediction entropy and ECE require observing test data. This section reports entropy and ECE changes for completeness, noting that they are not available as pre-deployment diagnostics.

Counter-intuitively, all three catastrophic tasks (coverage drop > 70%) show *decreasing* prediction entropy ( $\Delta H < 0$ ): the model becomes *more* confident at test time, not less. This occurs because concentrated dependence on an OOD feature produces arbitrary but confident predictions. In contrast, the two robust tasks with non-trivial drops (i-plant, i-shippoint; coverage drops of 10–19%) show *increasing* entropy ( $\Delta H > 0$ ), consistent with standard distributional broadening. A practitioner monitoring only entropy would correctly detect shift in moderate cases but would be *misled* in the most damaging catastrophic cases—observing decreasing uncertainty while coverage collapses. ECE increases across all tasks but does not discriminate between severity levels (both catastrophic and robust tasks show increases of +0.07 to +0.63). SHAP concentration remains the only evaluated metric available before deployment that correctly identifies all three catastrophic tasks.

Table 10: Concentration Check Applied to All Multiclass Tasks.

| Task                       | Conc. (%) | Step 2 | Actual | Note                       |
|----------------------------|-----------|--------|--------|----------------------------|
| <i>SALT (in-sample)</i>    |           |        |        |                            |
| s-payterms                 | 54.2      | VULN   | SEV    | —                          |
| s-shipcond                 | 50.7      | VULN   | SEV    | —                          |
| i-shippoint                | 48.8      | VULN   | ROB*   | High variance <sup>†</sup> |
| s-group                    | 47.3      | VULN   | SEV*   | —                          |
| s-office                   | 42.6      | VULN   | ROB    | Protective <sup>‡</sup>    |
| i-incoterms                | 28.9      | ROB    | ROB    | —                          |
| i-plant                    | 23.9      | ROB    | ROB    | —                          |
| s-incoterms                | 23.7      | ROB    | ROB    | —                          |
| <i>External (held-out)</i> |           |        |        |                            |
| Covertime                  | 49.9      | VULN   | SEV    | 83pp drop                  |
| KDDCup99                   | 18.7      | ROB    | ROB    | 11 classes                 |
| Gas Sensor                 | 7.6       | ROB    | ROB    | Vacuous sets               |

<sup>†</sup>Median drop 1.2%; most seeds maintain coverage. <sup>‡</sup>SALESORGANIZATION: train-validation Jaccard=0.61, importance>15%. In-sample F1 = 0.91 (1 FP: s-office). External: 3/3 correct (including hard direction).

Table 11: Pre-Deployment (SHAP) vs Post-Hoc (Entropy, ECE) Shift Diagnostics (10-seed ACI experiments). SHAP concentration is computed on validation data only.  $\Delta$ Entropy and  $\Delta$ ECE require test-time observations.

| Task                | Cat | SHAP Conc. <sup>†</sup> | $\Delta$ Entropy | $\Delta$ ECE |
|---------------------|-----|-------------------------|------------------|--------------|
| <i>Severe tasks</i> |     |                         |                  |              |
| s-shipcond          | SEV | 50.7%                   | −0.746           | +0.408       |
| s-group             | SEV | 47.3%                   | −1.274           | +0.176       |
| s-payterms          | SEV | 54.2%                   | −1.979           | +0.629       |
| <i>Robust tasks</i> |     |                         |                  |              |
| i-plant             | ROB | 23.9%                   | +0.181           | +0.1314      |
| i-shippoint         | ROB | 48.8%                   | +0.469           | +0.0727      |
| s-incoterms         | ROB | 23.7%                   | +0.364           | +0.350       |
| s-office            | ROB | 42.6%                   | +0.032           | +0.104       |

<sup>†</sup>Pre-deployment diagnostic (computed on validation data only). item-incoterms omitted (3-seed pilot only).

## G TASK INDEPENDENCE ANALYSIS

We analyze pseudo-replication in the 50-seed design using intraclass correlation (ICC).

**Between-task ICC.** For coverage drops,  $\text{ICC} = 0.675$  (95% CI: [0.47, 0.90];  $F(7, 392) = 104.6$ ), confirming that task identity explains the majority of seed-level variance. The design effect is 34.1, yielding effective  $n_{\text{eff}} = 400/34.1 = 11.7$ —exceeding the 8 task-level observations. For test coverages,  $\text{ICC} = 0.720$  (95% CI: [0.52, 0.92]). These values confirm that the  $n = 8$  tasks constitute meaningfully independent observations for the Spearman correlation analysis.

**Per-task ICC(val,test).** For each task, we compute ICC between the 50 validation coverages and 50 test coverages (paired by seed). All 8 per-task ICCs are negative (−0.80 to −0.10), indicating that seeds producing above-average validation coverage do *not* produce above-average test coverage. This means paired Wilcoxon tests are conservative: the effective sample size for within-task paired tests *exceeds* 50, so the reported  $p$ -values are, if anything, too large.

**Multiple comparisons.** We tested 5 concentration metrics (top-1, top-2, top-3, HHI, entropy) against coverage drop. Under Holm–Bonferroni correction for this 5-test family, the adjusted  $p$ -value for top-1 concentration is  $5 \times 0.010 = 0.050$ , at the conventional significance boundary. For the 3 correlation subsets ( $n = 8, n = 9, n = 11$ ), all adjusted  $p$ -values remain significant (Holm-corrected  $p \leq 0.021$ ). We note that top-1 was the a priori natural choice (Section 3.6), not one of 5 equally plausible alternatives; the correction is therefore conservative.

Table 12: Per-Task ICC Between Validation and Test Coverages (50 Seeds).

| Task        | ICC(val,test) | Design Effect | Eff. $n$ |
|-------------|---------------|---------------|----------|
| s-payterms  | −0.80         | 0.20          | 250      |
| s-shipcond  | −0.77         | 0.23          | 220      |
| s-group     | −0.72         | 0.28          | 177      |
| i-plant     | −0.42         | 0.58          | 86       |
| i-incoterms | −0.37         | 0.63          | 80       |
| s-incoterms | −0.35         | 0.65          | 77       |
| s-office    | −0.15         | 0.85          | 59       |
| i-shippoint | −0.10         | 0.90          | 55       |

## H CONFORMITY SCORE STOCHASTIC DOMINANCE

Table 13 reports two-sample KS statistics comparing calibration and test APS conformity scores (seed=42, non-randomized APS). Higher conformity scores indicate the model requires more cumulative probability mass to reach the true label—i.e., the model is less confident about the correct class.

Table 13: KS Tests: Calibration vs. Test Conformity Score Distributions (seed=42).

| Task                                                     | KS    | $p$          | Mean <sub>cal</sub> | Mean <sub>test</sub> |
|----------------------------------------------------------|-------|--------------|---------------------|----------------------|
| <i>Catastrophic tasks (test scores &gt; calibration)</i> |       |              |                     |                      |
| s-shipcond                                               | 0.956 | $< 10^{-10}$ | 0.54                | 0.98                 |
| s-payterms                                               | 0.748 | $< 10^{-10}$ | 0.65                | 0.92                 |
| s-group                                                  | 0.676 | $< 10^{-10}$ | 0.98                | 1.00                 |
| <i>Robust tasks</i>                                      |       |              |                     |                      |
| i-plant                                                  | 0.741 | $< 10^{-10}$ | 0.73                | 0.86                 |
| s-incoterms                                              | 0.633 | $< 10^{-10}$ | 0.75                | 0.87                 |
| s-office                                                 | 0.994 | $< 10^{-10}$ | 1.00                | 1.00                 |
| i-incoterms                                              | 0.559 | $< 10^{-10}$ | 0.78                | 0.66                 |
| i-shippoint                                              | 0.505 | $< 10^{-10}$ | 0.71                | 0.58                 |

All KS tests are two-sided. Non-randomized APS scores (conservative coverage guarantee). For catastrophic tasks, test scores shift rightward (stochastic dominance), consistent with the heuristic argument in Section 4. Two robust tasks (i-incoterms, i-shippoint) show test scores shifted *leftward*—the model becomes more confident during COVID, possibly due to reduced product diversity.

## I RAPS COMPARISON

Table 14 compares APS and RAPS coverage drops across all 8 tasks (single seed,  $\lambda = 0.01$ ,  $k_{\text{reg}} = 5$ ).

Table 14: APS vs. RAPS Coverage Drop (single seed). RAPS uses  $\lambda = 0.01$ ,  $k_{\text{reg}} = 5$ . Tasks ordered by APS drop magnitude.

| Task        | Cl  | APS Drop | RAPS Drop | $\Delta$ | Note                   |
|-------------|-----|----------|-----------|----------|------------------------|
| s-shipcond  | 45  | 92.3%    | 92.3%     | 0.0      | Identical              |
| s-group     | 462 | 67.3%    | 10.8%     | −56.5    | Regularization helps   |
| i-incoterms | 13  | 19.8%    | 19.8%     | 0.0      | Identical              |
| i-plant     | 35  | 16.7%    | 17.1%     | +0.4     | Identical              |
| i-shippoint | 70  | 2.9%     | 6.3%      | +3.4     | Small change           |
| s-incoterms | 13  | 3.4%     | 3.4%      | 0.0      | Identical              |
| s-payterms  | 135 | −8.0%    | 13.5%     | +21.5    | Regularization changes |
| s-office    | 25  | 0.1%     | 0.1%      | 0.0      | Identical              |

APS-vs-RAPS drop correlation:  $\rho = 0.619$ ,  $p = 0.10$ . Single-seed drops differ slightly from 50-seed means in Table 1. For sales-payterms, APS shows −8.0% (overcoverage) in single seed while 50-seed mean shows 77.1% drop, indicating high seed sensitivity for this task.

## J MODEL CLASS SENSITIVITY

To test whether SHAP concentration is diagnostic across model classes, we replicate the analysis with `RandomForestClassifier` (500 trees), `XGBClassifier`, and `CatBoostClassifier` (ordered boosting), all with `seed=42` and the same data pipeline (Table 15).

Table 15: SHAP Concentration and Coverage Drop Across Model Classes.

| Task        | LGB               | RF   | XGB  | CB   | LGB               | RF   | XGB  | CB   |
|-------------|-------------------|------|------|------|-------------------|------|------|------|
|             | Concentration (%) |      |      |      | Coverage Drop (%) |      |      |      |
| s-shipcond  | 50.7              | 57.4 | 44.6 | 53.7 | 71.6              | 15.8 | 47.2 | 1.1  |
| s-group     | 47.3              | 22.5 | 33.3 | 31.8 | 71.2              | 50.0 | 56.8 | 14.5 |
| s-payterms  | 54.2              | 27.9 | 30.6 | 25.7 | 77.1              | 13.8 | 20.4 | -5.3 |
| i-shippoint | 48.8              | 52.1 | 41.0 | 43.1 | 18.5              | 22.7 | 80.4 | 61.8 |
| i-plant     | 23.9              | 26.8 | 26.0 | 29.0 | 10.6              | 15.3 | 9.7  | 8.5  |
| s-incoterms | 23.7              | 26.8 | 21.6 | 28.2 | 8.5               | 2.5  | 3.7  | -1.5 |
| i-incoterms | 28.9              | 17.5 | 18.4 | 21.7 | 11.3              | 2.9  | 1.1  | -0.4 |
| s-office    | 42.6              | 24.5 | 36.8 | 33.5 | 0.1               | 0.0  | 0.0  | -0.0 |

**Results.** The diagnostic’s predictive power varies by model class (Table 16). All three gradient-boosted implementations show positive correlation: LightGBM ( $\rho = 0.833$ ,  $p = 0.010$ ), CatBoost ( $\rho = 0.667$ ,  $p = 0.071$ ), and XGBoost ( $\rho = 0.548$ ,  $p = 0.16$ ). RandomForest ( $\rho = 0.30$ ,  $p = 0.47$ ) does not replicate.

**Mixed-effects analysis.** Since (task, model) pairs share data and temporal split, we fit a linear mixed-effects model with task as a random intercept to properly account for clustering:  $\text{Drop}_{ij} = \beta_0 + \beta_1 \cdot \text{Concentration}_{ij} + u_i + \varepsilon_{ij}$ , where  $u_i \sim \mathcal{N}(0, \sigma_u^2)$  is the task random effect. Across the three gradient-boosted models ( $n = 24$ , 8 tasks  $\times$  3 models), the concentration effect is  $\beta_1 = 1.64$  (95% CI [0.70, 2.58],  $p = 0.0006$ ): each 1% increase in concentration predicts 1.6 pp additional coverage drop, after accounting for task-level clustering. Including model class as a fixed effect yields  $\beta_1 = 1.66$  ( $p = 0.001$ ). With all four models including RF ( $n = 32$ ):  $\beta_1 = 1.06$  ( $p = 0.010$ ).

Table 16: Concentration–Drop Relationship by Model Class.

| Model                                  | Method               | Conc.–Drop $\rho$ | $p$          |
|----------------------------------------|----------------------|-------------------|--------------|
| LightGBM                               | Boosting (GOSS)      | <b>0.833</b>      | <b>0.010</b> |
| CatBoost                               | Boosting (ordered)   | 0.667             | 0.071        |
| XGBoost                                | Boosting (histogram) | 0.548             | 0.16         |
| RandomForest                           | Bagging              | 0.30              | 0.47         |
| Mixed-effects, 3 boosting ( $n = 24$ ) |                      | $\beta_1 = 1.64$  | <b>0.001</b> |
| Mixed-effects, all 4 ( $n = 32$ )      |                      | $\beta_1 = 1.06$  | <b>0.010</b> |

**Interpretation.** All three gradient-boosted implementations show the concentration–drop relationship, but with different vulnerability profiles. CatBoost’s ordered boosting produces dramatically different task-level vulnerabilities: LGB-catastrophic tasks (s-shipcond: 71.6% LGB drop) show only 1.1% CatBoost drop, while i-shippoint reverses (18.5% LGB  $\rightarrow$  61.8% CatBoost). This confirms each model’s SHAP concentration captures its *own* learned dependence—not a universal data property—and that cross-model transfer is unreliable (LGB conc.  $\rightarrow$  CatBoost drop:  $\rho = 0.07$ ). RF’s bagging produces smoother probability surfaces, compressing both concentration range and vulnerability profiles.

The diagnostic is *model-specific*: it measures how a particular model’s learned dependence structure creates vulnerability. Practitioners should compute concentration for their *deployed* model class rather than transferring results across model families.